Course Code: EET 305

Course Name: SIGNALS AND SYSTEMS

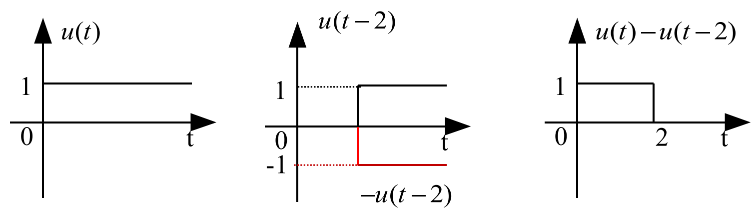
Max. Marks: 100 Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks) Marks

1 Non- linear, causal, time variant 3

2 3



3 System should be absolutely Integraable, minimum number of maxima and minima 3

4 Perceval's theorem state that energy of continuous time aperiodic signal is equal to summation of corresponding continuous time Fourier transform representation. 3

5 Positive real functions are property of Laplace transform that causal signal always lie in positive side of imaginary axis. 3

6 A first order control system is defined as a type of control system whose input-output relationship (also known as a transfer function) is a first-order differential equation 3

7 Transfer function = $\frac{1-e^{-sT}}{s}$ 3

8 Nyquist rate = 200Hz 3

Nyquist width = 5ms

9 Reversal property of DTFS states that the scaled version of input signal. That is the time reversal property of Fourier series coefficients is time reversal of the corresponding sequence of Fourier series. 3

10 $X[\omega]=\frac{1}{\left(1-ae^{-j\omega}\right)}$ 3

PART B

(Answer one full question from each module, each question carries 14 marks)

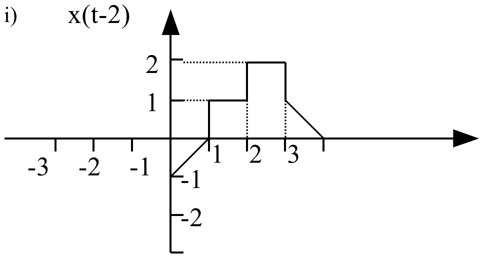
Module -1

11 a) Linear means something related to a line. All the linear equations are used to construct a line. A non-linear equation is such which does not form a straight line. It looks like a curve in a graph and has a variable slope value. If the relationship between y and x is linear (straight line) and crossing through origin then the system is linear. 7

b) 7

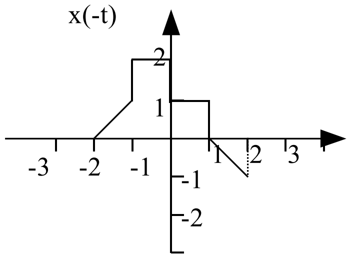
$$y(t)=\begin{cases} 0;t<3 \\ \dfrac{1-e^{-3(t-3)}}{3};3\le t\le 5 \\ \dfrac{e^{-3(t-5)}-e^{-3(t-3)}}{3};t\ge 5 \end{cases}$$

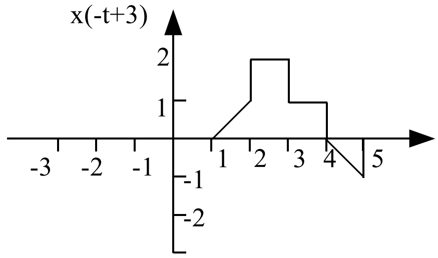
12 a) i) 7

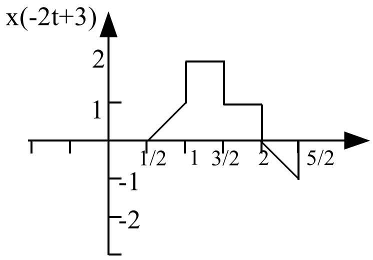


(3 marks)

ii)







(4 marks)

b) Energy signal is a signal whose energy is finite and power is zero whereas Power signal is a signal whose power is finite and energy is infinite. where E is the energy of the signal and P is the power of the signal. 7

Energy is infinity. Power = 0.5 W. Hence power signal

Module -2

13 a) 7

$$a_o = \frac{A}{\pi}$$

$$a_n = \frac{2A}{\pi(1-n^2)} \quad n \neq 1;\ a_1 = 0$$

$$\left.\begin{aligned} b_n &= 0;\ n \neq 1 \\ b_1 &= \frac{A}{2} \end{aligned}\right\}$$

$$x(t)=\frac{A}{\pi}+\frac{A}{2}\sin t+\sum_{n=2}^{\infty}\frac{2A}{\pi(1-n^2)}\cos nt$$

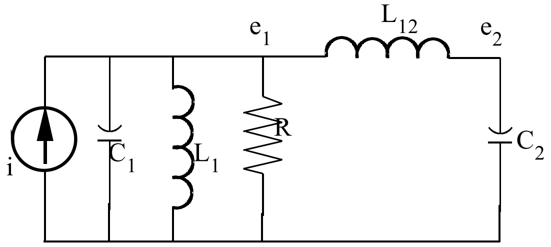
b) $$X(j\omega)=\frac{j\omega}{\omega_o^2-\omega^2}+\frac{1}{2}\pi\delta(\omega-\omega_o)+\frac{1}{2}\pi\delta(\omega+\omega_o)$$ 7

14 a) $$M_1\frac{d^2y_1}{dt^2}+f\frac{dy_1}{dt}+K_1y_1+K_{12}(y_1-y_2)=F(t)$$ 8

$$M_2\frac{d^2y_2}{dt^2}+K_{12}(y_2-y_1)=0$$

Differential equations

Force current analogous circuit



b) Transfer function, $H(s)=\frac{Y(s)}{X(s)}=\frac{1}{s^2+s-2}$ 6

Impulse response , $h(t)=L^{-1}\left[H(s)\right]=L^{-1}\left[\frac{1}{s^2+s-2}\right]=\frac{1}{3}\left[e^{t}-e^{-2t}\right]$

Module -3

15 a) Forward Paths: 14

$$\left.\begin{array}{l}P_1=G_2G_4G_6\,;P_2=G_3G_5G_7\,;P_3=G_2G_1G_7\,;P_4=G_3G_8G_6\\ P_5=G_3G_8H_1G_1G_7;P_6=G_2G_1H_2G_8G_6\end{array}\right\}\text{ 6 marks}$$

Loops:

$L_{11}=G_4H_1\,;L_{21}=G_5H_2\,;L_{31}=G_8H_1G_1H_2$ - 3 marks

Two non- touching loops –

$L_{21}=G_4G_5H_1H_2$ - 1 mark

Determinant –

$\Delta=1-\left(G_4H_1+G_5H_2+G_1G_8H_1H_2\right)+G_4G_5H_1H_2$ - 1 mark

Mason’s gain formula

Transfer function

$$\frac{C(s)}{R(s)} = \frac{G_2G_4G_6(1-G_5H_2)+G_3G_5G_7(1-G_4H_1)+G_1G_2G_7+G_3G_6G_8}{1-\left(G_4H_1+G_5H_2+G_1G_8H_1H_2\right)+G_4G_5H_1H_2}$$ ---- 2 marks

b)

16 a) The case with the Dirac delta for the input, a LTI system is causal if and only if: **h(t) = L(δ(t)) = 0, t < 0**. In other words the impulse response of a LTI system has to be zero for negative time for the system to be causal. 7

b) Routh array 7

s^5	1	2	11
s^4	2	4	10
s^3	0(ε)	6	
s^2	$\frac{4\varepsilon-12}{\varepsilon}$	10	
s^1	6	0	
s^0	10		

Stability analysis-Two sign changes. Hence unstable.

Module -4

17 a) $x[n]=\frac{1}{9}\left[2(1)^n+7(-2)^n\right]u[n]$ 7

b) $y(n)=\frac{1}{3}\left(\frac{1}{2}\right)^n\left[1+2(-2)^n\right]u(n)$ 7

18 a) [x1(n)* x2(n)]*h(n)= x1(n)* [x2(n)*h(n)], x1(n) and x2(n) are inputs and h(n) is the impulse response. This can be proved by considering two x1(n)* x2(n) as one output and then using the commutative property proof. 7

b) $X[z]=\frac{\left[z^2-8z+7\right]}{z^7(z-1)^2}$ 7

Module -5

19 a) Impulse response: $h[n]=\frac{6}{5}\left[\left(\frac{1}{2}\right)^n-\left(\frac{1}{3}\right)^n\right]u[n]$ 14

Frequency response:

$$\left|H(e^{j\omega})\right|=\frac{1}{\left[\left\{\left(\cos\omega-\frac{1}{2}\right)^2+\sin^2\omega\right\}\left\{\left(\cos\omega+\frac{1}{3}\right)^2+\sin^2\omega\right\}\right]^{\frac{1}{2}}}$$

$$\angle H\left(e^{j\omega}\right) = \omega = \tan^{-1}\frac{\sin\omega}{\left(\cos\omega - \frac{1}{2}\right)} - \tan^{-1}\frac{\sin\omega}{\left(\cos\omega + \frac{1}{3}\right)}$$

b)

20 a) Jury's Conditions 6

Row	z^0	z^1	z^2	z^3
1	0.05	-0.25	0.2	1
2	1	0.2	-0.25	0.05
3	-0.9975	-0.2125	0.26	

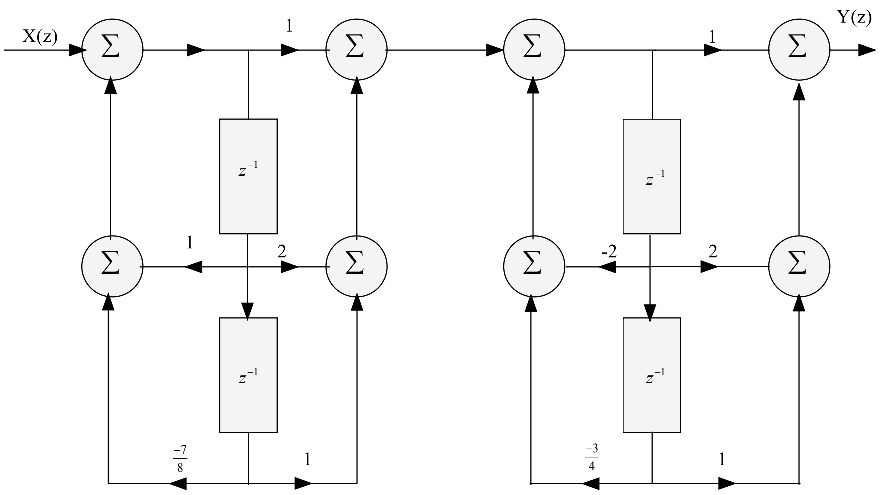
$|a_0| < |a_3| \Rightarrow |0.05| < |1|$ Satisfied

$|b_0| > |b_2| \Rightarrow |-0.9975| > |0.26|$ Satisfied

Stable

b) Realization 8

$$H(z) = \frac{\left(1+2z^{-1}+z^{-2}\right)}{\left(1-z^{-1}+\frac{7}{8}z^{-2}\right)} \cdot \frac{\left(1+2z^{-1}+z^{-2}\right)}{\left(1+2z^{-1}+\frac{3}{4}z^{-2}\right)}$$



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